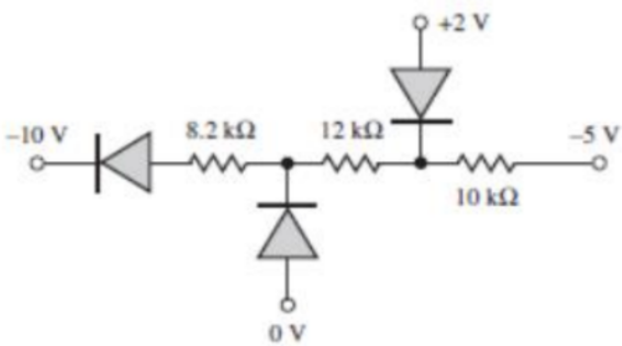
Diode DC Analysis
Practice Problems Before Tutorial

Dear Students,
You all are instructed to attempt to solve these problems before joining the upcoming tutorial session on 13th February, 2024 (Tuesday).
N.B.: *Every student joining the tutorial session must attempt the problems before joining.*

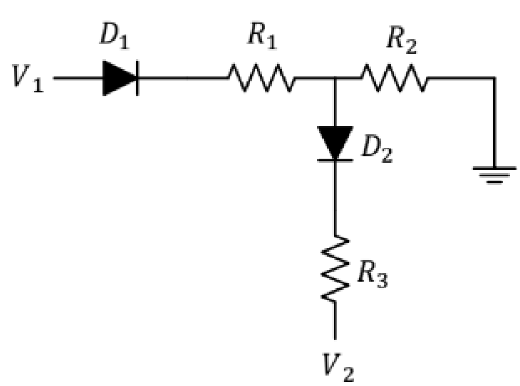
Problem 1: Determine the following for the circuits shown in figures below:
a. The conduction states (ON/OFF) of the diodes.
b. The diode currents.
Consider all the diodes are Si diodes. Solve the problem using **both** ON and OFF assumptions.



Problem 2: Solve the circuit making the following assumptions and justify the assumptions made.

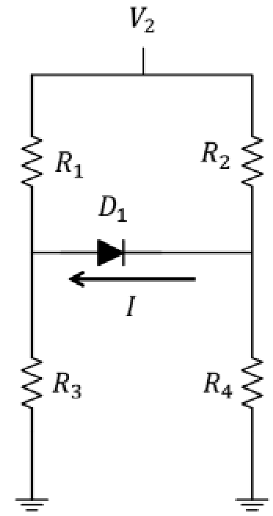
D1	D2
ON	OFF
OFF	ON

Consider Si diodes. R1 = 2k, R2 = 0.5k, R3 = 5k, V1 = 4V and V2 = 2V



Problem 3: i. Determine the conduction state of the diode in each circuit using
a. on assumption,
b. off assumption.

ii. Determine I. consider Si diode. R1 = 2k, R2 = 0.5k, R3 = 5k, R4 = 2k, V2 = 2V



Problem 4: i. Determine the conduction states (On or Off) of all the diodes.
ii. Determine the currents through the diodes.
Consider D1 & D2: Si and D3 & D4: Ge . R1 = 2k, R2 = 5k, R3 = 7.2k, for V1 = −5V and 5V.